

Dual Sharing Of Real Time Clock – An Example

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I have an old window AC in my lab, which is at the terrace. The AC needs to be run intermittently, especially when I am out of station and the lab PC needs to be kept cool during noon time when the temperature shoots to 35 + degree centigrade, irrespective of summer or winter. Of course, the AC's thermostat works, but it needs to be adjusted continuously—else it will keep running for prolonged periods during summers, which I want to avoid.

The following timer based on Arduino does exactly that. I can set the on-time and off-time based on the entire calendar timings! It is based on a simple DS3231 real time clock (RTC) timer, an ATMEGA328 Arduino chip, and a small 5V relay to run the 5kW contactor that controls the AC.

Trigger to go further

“Uncle you have made a whole calendar-dependent timer, but you are not showing clock/calendar, or the alarm time, etc, anywhere. It should display time at least to know what it's going

to do next, or what it has done so far,” my friend's son Reon declared as his first impression.

The whole can of worms got spilled out when I failed to comply miserably to his request! For running the on-time and off-time, I had used switch statements combined with ‘if’ statements in *DS3231_AC_Scheduler.ino* Arduino code. Therefore, the clock necessarily stopped inside the ‘if’ loop. Only when the ‘if’ statement expired the loop ended, the clock started, and the next ‘switch’ statement was executed. Therefore, the clock display ran intermittently with this sketch when it executed the loop. Thus, I failed to achieve the continuous display of the clock, as Reon desired!

Alas! If I could run a second Arduino independently, on same time-sharing clock, it would solve the problem fully. Else, can I put a second clock in parallel to this Arduino?

Idea

Man with one clock always knows the time, but with two clocks he's

never quite sure! Therefore, when it's a time-dependent project, it's always better to have a single clock shared by all. Though I didn't go through too much pain to make it super-accurate, it passes the eyeball test quite well, and that's good enough for me; I'm not managing satellites or cell towers.

A multi-master I2C RTC set-up can be built using RTC DS3231. It is a pretty precise RTU, which has an inbuilt temperature sensor for adjusting the clock frequencies for temperature corrections. It maintains very precise timings over long duration.

The principle of operation

The DS3231 is a very precise RTU, which is being shared here by two Arduinos, and can be shared by more microcontrollers as well. One Arduino runs the timer while the other displays the time and all other allied information to be shown. For handling hour, minute, and second in simple decimal way, I followed a trick that is also used by many on the Internet:

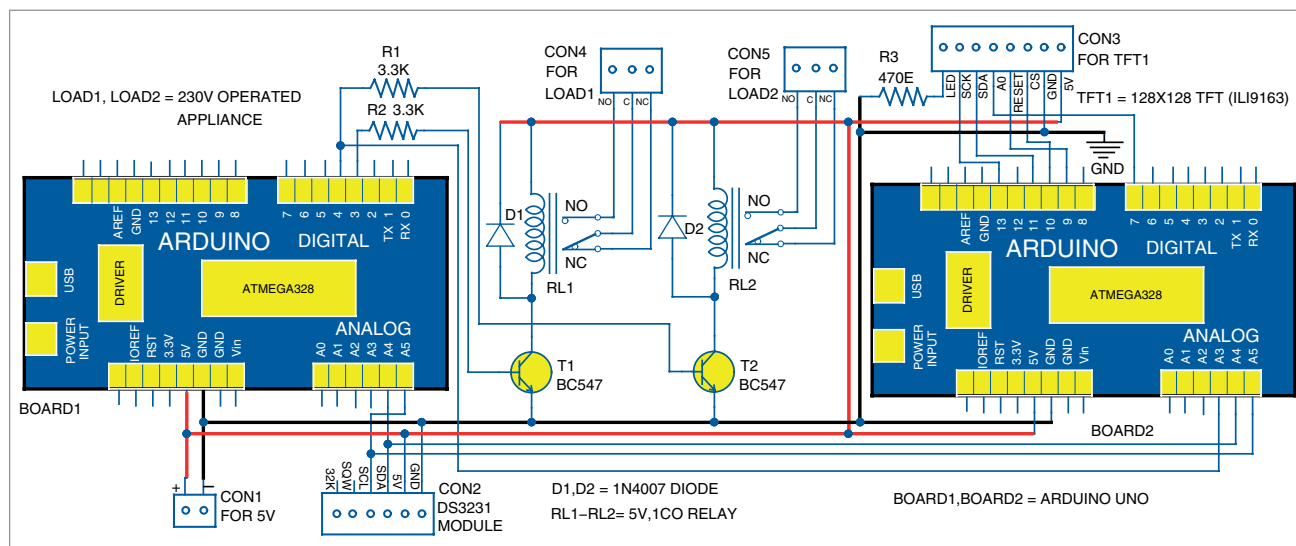


Fig. 1: Circuit diagram of dual sharing RTC